# Nick Choksi

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## Appointments

2025-Present **51 Pegasi b Postdoctoral Fellow**, *Caltech*.
Host: Konstantin Batygin

## Education

2019-2025 **Ph.D. in Astrophysics**, *University of California*, *Berkeley*.
Advisor: Eugene Chiang

2019 **B.A. in Physics & Astrophysics**, *University of California*, *Berkeley*,
Advisor: Oleg Gnedin, Marta Volonteri, Peter Behroozi.

## Research interests

theoretical astrophysics, planet formation, gravitational dynamics, circumstellar accretion disks, stellar cluster formation

## Publications

13 lead-authored publications with 428 citations and $h = 10$
18 total publications with 805 citations and $h = 14$

1. **Choksi**, Lithwick, Chiang, & Li, "Two-stage disruption of resonant chains," MNRAS in press, arXiv 2604.05035.
2. **Choksi** & Chiang, "Spectral Energy Distributions of Disc-Embedded Accreting Protoplanets," MNRAS 537 2945 (2025), arXiv 2403.10057.
3. Dai, Goldberg, Batygin, van Saders, Chiang, **Choksi**, et al., "The Prevalence of Resonance Among Young, Close-in Planets," ApJ 168, 239 (2024), arXiv 2406.06885.
4. Li, Chiang, **Choksi**, and Dai, "The Resonant Remains of Broken Chains from Major and Minor Mergers," ApJ in press, arXiv 2408.10206.
5. **Choksi**, Chiang, Fung, & Zhu, "The maximum accretion rate of a protoplanet: how fast can runaway be?," MNRAS 525, 2806 (2023), arXiv 2305.01684.
6. **Choksi** & Chiang, "Exciting the TTV Phases of Resonant Sub-Neptunes," MNRAS 522, 1914 (2023), arXiv 2211.15701.
7. Rein & **Choksi**, "An Implementation of Stochastic Forces for the N-body Code `REBOUND`," RNAAS 6, 5 (2022), arXiv 2205.06757.
8. **Choksi** & Chiang, "Testing planet formation from the ultraviolet to the millimeter," MNRAS 510, 1657 (2021), arXiv 2110.00029.
9. **Choksi**, Chiang, Connolly, Gainsforth, and Westphal, "Chondrules from high-velocity collisions: thermal histories and the agglomeration problem," MNRAS 503, 3297 (2021), arXiv 2009.10093.
10. **Choksi** & Chiang, "Sub-Neptune Formation: The View from Resonant Planets,"

MNRAS 495, 4192 (2020), arXiv 2003.03388.

11. **Choksi** & Kruijssen, "On the initial mass-radius relation of stellar clusters," MNRAS 507, 5492, arXiv 1912.05560.
12. **Choksi** & Gnedin, "Origins of scaling relations of globular cluster systems," MNRAS 488, 5409 (2019), arXiv 1905.05199.
13. **Choksi** & Gnedin, "Formation of Globular Cluster Systems II: Impact of the cutoff of the cluster initial mass function," MNRAS 486, 331 (2019), arXiv 1810.01888.
14. **Choksi**, Volonteri, Colpi, Gnedin, and Li, "The star clusters that make black hole binaries across cosmic time," ApJ 873, 100 (2019), arXiv 1809.01164.
15. El-Badry, Quataert, Weisz, **Choksi**, and Boylan-Kolchin, "The formation and hierarchical assembly of globular cluster populations," MNRAS 482, 4528 (2018), arXiv 1805.03652.
16. **Choksi**, Gnedin, and Li, "Formation of globular cluster systems: from dwarf galaxies to giants," MNRAS 480, 2343 (2018), arXiv 1801.03515.
17. **Choksi**, Behroozi, Volonteri, Schneider, Ma, Silk, and Moster, "Recoiling supermassive black hole escape velocities from dark matter halos," MNRAS 472, 1526 (2017), arXiv 1707.06220.

## PI/co-PI telescope proposals

1. *Measuring the composition and circumplanetary environment of a gap-clearing protoplanet,* James Webb Space Telescope Cycle 5 GO 9993, 44 hours.

## Invited Seminars

1. Colloquium, UCLA, 2026
2. Distinguished Seminar Series, Los Alamos National Lab, 2026
3. Theory Seminar, CIERA, 2025
4. Exoplanet Seminar, Princeton, 2024
5. TAPIR Seminar, Caltech, 2024
6. Trottier Space Sciences Seminar, McGill, 2023
7. Star and Planet Formation Seminar, Hawaii IfA, 2023
8. CfA Seminar, Harvard, 2023

## Conference talks

1. International Conference on Exoplanets and Planet Formation, Shanghai, 2025
2. Gas Accretion in Planet Formation, Heidelberg, 2025
3. Exoplanets V, Leiden, 2024 (**two** accepted talks)
4. Open problems in the astrophysics of gas giants, Patagonia, 2023
5. Disk hydrodynamics and planet formation, Tucson, 2023
6. Other Worlds Laboratory Summer Program, Santa Cruz, 2023
7. Bay Area Exoplanets Meeting, Santa Cruz, 2023
8. Exoplanet Demographics, 2020
9. Bay Area Exoplanets Meeting, 2020
10. Formation of stars and massive clusters in dwarf galaxies over cosmic time, Leiden,

2019 (**invited**)

11. Formation of globular clusters at high and low-$z$, Sesto, 2018
12. Galaxy formation workshop, Santa Cruz, 2017
13. Massive black holes in evolving galaxies, Institut d'Astrophysique Paris, 2017

## Honors & Awards

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| 2025 | Mary Elizabeth Uhl Award, Berkeley Astronomy |
| 2024 | Robert J. Trumpler Award, Berkeley Astronomy |
| 2024 | UC Dissertation-Year Fellow ($40,000) |
| 2019-2024 | NSF Graduate Research Fellowship |
| 2020 | Esper Larsen Jr. Grant, Berkeley Earth & Planetary Science Department ($20,000) |
| 2021 | H2H8 Fellow ($10,000) |
| 2019 | Student commencement speaker, Berkeley Astronomy |
| 2019 | Finalist, Hertz Fellowship |
| 2018 | Isidore Pomerantz Award, Berkeley Physics |

## Service, Outreach, and Teaching

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| Current | **Referee**, *MNRAS, ApJ, Icarus*. |
| 2021 | **Team member, Berkeley Discover Astronomy & Physics**. Collaborator on successful proposal ($800,000) to revamp undergraduate teaching in physics & astronomy at Berkeley. |
| 5 semesters | **Graduate student instructor**, *Astrophysics I & II, Astronomy for non-majors*. |